

HW12

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Exercise 12.1

设二维随机变量 (X, Y) 的联合分布列为

	Y		
X	-1	0	1
0	0.07	0.18	0.15
1	0.08	0.32	0.20

试求 X^2 与 Y^2 的协方差.

Solution. $\text{Cov}(X^2, Y^2) = E[X^2 Y^2] - E[X^2]E[Y^2]$

$$\begin{aligned} E[X^2] &= 0^2 \cdot P(X=0) + 1^2 \cdot P(X=1) \\ &= 0.08 + 0.32 + 0.20 = 0.60, \end{aligned}$$

$$\begin{aligned} E[Y^2] &= (-1)^2 \cdot P(Y=-1) + 0^2 \cdot P(Y=0) + 1^2 \cdot P(Y=1) \\ &= (0.07 + 0.08) + (0.15 + 0.20) = 0.50, \end{aligned}$$

$$\begin{aligned} E[X^2 Y^2] &= \sum_{i,j} x_i^2 y_j^2 p_{ij} \\ &= 1^2 \cdot (-1)^2 \cdot 0.08 + 1^2 \cdot 1^2 \cdot 0.20 = 0.28. \end{aligned}$$

代入公式得:

$$\text{Cov}(X^2, Y^2) = 0.28 - 0.60 \times 0.50 = -0.02.$$

◇

Exercise 12.2

把一颗骰子独立地掷 n 次, 求 1 点出现次数与 6 点出现次数的协方差及相关系数.

Proof. 设 $Z_1, \dots, Z_n$ 分别代表 n 枚骰子的点数, 定义

$$X_i = \begin{cases} 1, & Z_i = 1 \\ 0, & \text{其他} \end{cases}, \quad Y_i = \begin{cases} 1, & Z_i = 6 \\ 0, & \text{其他} \end{cases}$$

设 X, Y 分别为 1, 6 出现的次数, 则

$$X = \sum_{i=1}^n X_i, \quad Y = \sum_{i=1}^n Y_i$$

由 Cov 的双线性,

$$\begin{aligned} \text{Cov}(X, Y) &= \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n Y_i\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, Y_j) \end{aligned}$$

$$\begin{aligned} \text{Cov}(X_i, Y_j) &= E(X_i Y_j) - E(X_i) E(Y_j) \\ &= P(Z_i = 1, Z_j = 6) - P(Z_i = 1) P(Z_j = 6) \\ &= \begin{cases} 0, & i \neq j \\ -\frac{1}{36}, & i = j \end{cases} \end{aligned}$$

因此

$$\text{Cov}(X, Y) = \sum_{i=1}^n \text{Cov}(X_i, Y_i) = -\frac{n}{36}$$

$$\begin{aligned} \text{Var}(X) &= \text{Var}\left(\sum_{i=1}^n X_i\right) = \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i\right) \\ &= \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) \end{aligned}$$

其中

$$\text{Cov}(X_i, X_j) = E(X_i X_j) - E(X_i) E(X_j) = \begin{cases} \frac{5}{36}, & i = j \\ 0, & i \neq j \end{cases}$$

因此

$$\text{Var}(X) = \sum_{i=1}^n \frac{5}{36} = \frac{5n}{36}$$

由对称性,

$$\text{Var}(Y) = \frac{5n}{36}$$

最终

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}} = \frac{-\frac{n}{36}}{\frac{5n}{36}} = -\frac{1}{5}$$

□

Exercise 12.3

掷一颗骰子两次, 求其点数之和与点数之差的协方差.

Proof. 设 X, Y 为两次骰子的点数, 则

$$\begin{aligned}\text{Cov}(X + Y, X - Y) &= \text{Cov}(X, X) - \text{Cov}(Y, Y) \\ &= \text{Var}(X) - \text{Var}(Y)\end{aligned}$$

由于 X, Y 同分布, $\text{Var}(X) = \text{Var}(Y)$, 因此

$$\text{Cov}(X + Y, X - Y) = 0$$

□

Exercise 12.4

某箱装 100 件产品, 其中一、二和三等品分别为 80, 10 和 10 件. 现从中随机取一件, 定义两个随机变量 X_1, X_2 如下

$$X_i = \begin{cases} 1, & \text{若抽到 } i \text{ 等品,} \\ 0, & \text{其他,} \end{cases} \quad i = 1, 2.$$

试求随机变量 X_1 和 X_2 的相关系数 $\text{Corr}(X_1, X_2)$.

Proof. 由题意知 $X_1 \sim B(1, 0.8)$ 和 $X_2 \sim B(1, 0.1)$. 它们的方差分别为

$$\text{Var}(X_1) = 0.8 \times (1 - 0.8) = 0.16, \quad \text{Var}(X_2) = 0.1 \times (1 - 0.1) = 0.09.$$

由于取一件产品不可能同时为一等品和二等品, 故 $X_1X_2 = 0$. 协方差为

$$\text{Cov}(X_1, X_2) = E[X_1X_2] - E[X_1]E[X_2] = 0 - 0.8 \times 0.1 = -0.08.$$

因此, 相关系数为

$$\text{Corr}(X_1, X_2) = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\text{Var}(X_1)}\sqrt{\text{Var}(X_2)}} = \frac{-0.08}{\sqrt{0.16} \times \sqrt{0.09}} = -\frac{2}{3}.$$

□

Exercise 12.5

将一枚硬币重复掷 n 次, 以 X 和 Y 分别表示正面向上和反面向上的次数, 试求 X 和 Y 的协方差及相关系数.

Solution. 注意到

$$X + Y = n$$

因此

$$\text{Cov}(X, Y) = \text{Cov}(X, n - X) = -\text{Cov}(X, X) = -\text{Var}(X)$$

设 $Z_1, \dots, Z_n$ 为 n 次硬币出现的结果, 定义

$$X_i = \begin{cases} 1, & Z_i \text{ 出现正面} \\ 0, & Z_i \text{ 出现反面} \end{cases}$$

则

$$X = \sum_{i=1}^n X_i$$

$$\text{Var}(X) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j)$$

其中

$$\text{Cov}(X_i, X_j) = \begin{cases} \frac{1}{4}, & i = j \\ 0, & i \neq j \end{cases}$$

于是

$$\text{Var}(X) = \sum_{i=1}^n \left(\frac{1}{4}\right) = \frac{n}{4}, \quad \text{Var}(Y) = \text{Var}(n - X) = (-1)^2 \text{Var}(X) = \frac{n}{4}$$

因此

$$\begin{aligned}\text{Cov}(X, Y) &= -\text{Var}(X) = -\frac{n}{4} \\ \text{Corr}(X, Y) &= \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{-\frac{n}{4}}{\sqrt{\frac{n}{4}}\sqrt{\frac{n}{4}}} = -1\end{aligned}$$

◇

Exercise 12.6

在一个有 n 个人参加的晚会上, 每个人带了一件礼物, 且假定各人带的礼物都不相同。晚会期间各人从放在一起的 n 件礼物中随机抽取一件, 试求抽中自己礼品的人数 X 的均值和方差。

Proof. 设 $Z_1, \dots, Z_n$ 分别代表 n 个人抽中的礼物, 其中若 $Z_i = k$, 我们说第 i 个人抽中了第 k 个人的礼物. 定义

$$X_i = \begin{cases} 1, & Z_i = i \\ 0, & \text{其它} \end{cases}$$

则

$$X = \sum_{i=1}^n X_i$$

于是

$$E(X) = \sum_{i=1}^n E(X_i) = \sum_{i=1}^n \left(1 \cdot \frac{1}{n}\right) = 1$$

$$\text{Var}(X) = \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j)$$

其中

$$\text{Cov}(X_i, X_j) = \begin{cases} \frac{n-1}{n^2}, & i = j \\ \frac{1}{n^2(n-1)}, & i \neq j \end{cases}$$

因此

$$\text{Var}(X) = n \frac{n-1}{n^2} + (n^2 - n) \frac{1}{n^2(n-1)} = 1$$

□

Exercise 12.7

已知随机变量 X 与 Y 的相关系数为 ρ , 求 $X_1 = aX + b$ 与 $Y_1 = cY + d$ 的相关系数, 其中 a, b, c, d 均为非零正常数.

Proof.

$$\begin{aligned}\text{Cov}(X_1, Y_1) &= \text{Cov}(aX + b, cY + d) \\ &= ac \text{Cov}(X, Y) + \text{Cov}(b, cY) + \text{Cov}(aX, d) + \text{Cov}(b, d)\end{aligned}$$

由于常数作为随机变量, 与任意其他随机变量是相独立的, 上述四项只有第一项非零, 因此

$$\text{Cov}(X_1, Y_1) = ac \text{Cov}(X, Y)$$

此外,

$$\text{Var}(X_1) = \text{Var}(aX + b) = a^2 \text{Var}(X), \quad \text{Var}(Y_1) = \text{Var}(cY + d) = c^2 \text{Var}(Y)$$

因此

$$\begin{aligned}\text{Corr}(X_1, Y_1) &= \frac{\text{Cov}(X_1, Y_1)}{\sqrt{\text{Var}(X_1)}\sqrt{\text{Var}(Y_1)}} \\ &= \frac{ac \text{Cov}(X, Y)}{a \sqrt{\text{Var}(X)} c \sqrt{\text{Var}(Y)}} \\ &= a^2 c^2 \text{Corr}(X, Y) \\ &= a^2 c^2 \rho\end{aligned}$$

□

Exercise 12.8

设随机变量 X_1 与 X_2 独立同分布, 其共同分布为 $N(\mu, \sigma^2)$. 试求 $Y = aX_1 + bX_2$ 与 $Z = aX_1 - bX_2$ 的相关系数, 其中 a 与 b 为非零常数.

Solution. 由独立性,

$$\text{Cov}(X_1, X_2) = 0$$

由 Cov 的双线性,

$$\begin{aligned}\text{Cov}(Y, Z) &= \text{Cov}(aX_1 + bX_2, aX_1 - bX_2) = \\ &= a^2 \text{Cov}(X_1, X_1) - b^2 \text{Cov}(X_2, X_2) \\ &= a^2 \text{Var}(X_1) - b^2 \text{Var}(X_2) \\ &= (a^2 - b^2) \sigma^2\end{aligned}$$

$$\begin{aligned}
\text{Var}(Y) &= \text{Cov}(Y, Y) \\
&= \text{Cov}(aX_1 + bX_2, aX_1 + bX_2) \\
&= a^2 \text{Cov}(X_1, X_1) + b^2 \text{Cov}(X_2, X_2) \\
&= (a^2 + b^2) \sigma^2
\end{aligned}$$

$$\begin{aligned}
\text{Var}(Z) &= \text{Cov}(Z, Z) \\
&= \text{Cov}(aX_1 - bX_2, aX_1 - bX_2) \\
&= a^2 \text{Cov}(X_1, X_1) + b^2 \text{Cov}(X_2, X_2) \\
&= (a^2 + b^2) \sigma^2
\end{aligned}$$

因此

$$\text{Corr}(Y, Z) = \frac{\text{Cov}(Y, Z)}{\sqrt{\text{Var}(Y) \text{Var}(Z)}} = \frac{(a^2 - b^2) \sigma^2}{(a^2 + b^2) \sigma^2} = \frac{a^2 - b^2}{a^2 + b^2}$$

◇

Exercise 12.9

设二维随机变量 (X, Y) 服从二维正态分布 $N(0, 0, 1, 1, \rho)$.

(1) 求 $E(\max\{X, Y\})$;

(2) 求 $X - Y$ 与 XY 的协方差及相关系数.

Solution. 1.

$$\max\{X, Y\} = \frac{X + Y + |X - Y|}{2}$$

令 $Z = X - Y$, 则 Z 服从正态分布,

$$E(Z) = E(X - Y) = E(X) - E(Y) = 0$$

$$\text{Var}(Z) = \text{Cov}(X - Y, X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y) = 2 - 2\rho$$

因此 $Z \sim N(0, 2 - 2\rho)$ 从而

$$\begin{aligned}
E(|Z|) &= 2 \int_0^\infty x \frac{1}{\sqrt{2\pi(2-2\rho)}} e^{-\frac{x^2}{4-4\rho}} dx \\
&= 2\sqrt{1-\rho} \int_0^\infty \frac{1}{\sqrt{\pi}} e^{-\left(\frac{x}{2\sqrt{1-\rho}}\right)^2} d\left(\frac{x}{2\sqrt{1-\rho}}\right)^2 \\
&= 2\sqrt{\frac{1-\rho}{\pi}}
\end{aligned}$$

故

$$E(\max\{X, Y\}) = \frac{1}{2}(E(X) + E(Y) + E(|Z|)) = \sqrt{\frac{1-\rho}{\pi}}$$

2. 令 $U = X - Y$, $V = XY$, 则 $E(U) = E(X) - E(Y) = 0$.

$$\begin{aligned}\text{Cov}(U, V) &= E(UV) - E(U)E(V) = E(UV) \\ &= E(X^2Y) - E(XY^2)\end{aligned}$$

由于 $(-X, -Y)$ 与 (X, Y) 同分布,

$$f_{X,Y}(x, y) = f_{-X,-Y}(x, y) = f_{X,Y}(-x, -y)$$

$$\begin{aligned}E(X^2Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 y f_{X,Y}(x, y) \, dx \, dy \\ &= \int_{-\infty}^0 \int_{-\infty}^{\infty} x^2 y f_{X,Y}(x, y) \, dx \, dy + \int_0^{\infty} \int_{-\infty}^{\infty} x^2 y f_{X,Y}(x, y) \, dx \, dy \\ &= - \int_0^{\infty} \int_{-\infty}^{\infty} x^2 y f_{X,Y}(x, y) \, dx \, dy + \int_0^{\infty} \int_{-\infty}^{\infty} x^2 y f_{X,Y}(x, y) \, dx \, dy \\ &= 0\end{aligned}$$

类似地,

$$E(XY^2) = 0$$

因此

$$\text{Cov}(U, V) = 0, \quad \text{Corr}(U, V) = 0$$

$X - Y$ 与 XY 的协方差和相关系数均为零.

◇

Exercise 12.10

设二维随机变量 X, Y 的联合密度函数为

$$p(x, y) = \begin{cases} \frac{6}{7} \left(x^2 + \frac{xy}{2} \right), & 0 < x < 1, 0 < y < 2, \\ 0, & \text{其他.} \end{cases}$$

求 X 与 Y 的协方差及相关系数。

Solution.

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

其中

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x p_X(x) dx = \int_{-\infty}^{\infty} x \int_{-\infty}^{\infty} p(x, y) dy dx \\ &= \int_0^1 \int_0^2 \frac{6}{7} \left(x^3 + \frac{1}{2} x^2 y \right) dy dx \\ &= \int_0^1 \left(\frac{12}{7} x^3 + \frac{6}{7} x^2 \right) dx \\ &= \frac{3}{7} + \frac{2}{7} = \frac{5}{7} \end{aligned}$$

$$\begin{aligned} E(Y) &= \int_{-\infty}^{\infty} y p_Y(y) dy = \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} p(x, y) dx dy \\ &= \int_0^2 \int_0^1 \frac{6}{7} \left(x^2 y + \frac{1}{2} x y^2 \right) dx dy \\ &= \int_0^2 \left(\frac{2}{7} y + \frac{3}{14} y^2 \right) dy \\ &= \frac{4}{7} + \frac{4}{7} = \frac{8}{7} \end{aligned}$$

$$\begin{aligned} E(XY) &= \int_{-\infty}^{\infty} x y p(x, y) dx dy \\ &= \int_0^2 \int_0^1 \frac{6}{7} \left(x^3 y + \frac{1}{2} x^2 y^2 \right) dx dy \\ &= \int_0^2 \left(\frac{3}{14} y + \frac{1}{7} y^2 \right) dy \\ &= \frac{3}{7} + \frac{8}{21} = \frac{17}{21} \end{aligned}$$

因此

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{17}{21} - \frac{5}{7} \cdot \frac{8}{7} = -\frac{1}{147}$$

$$\begin{aligned}
E(X^2) &= \int_{-\infty}^{\infty} x^2 \int_{-\infty}^{\infty} p(x, y) dy dx \\
&= \int_0^1 \int_0^2 \frac{6}{7} \left(x^4 + \frac{1}{2} x^3 y \right) dy dx \\
&= \int_0^1 \frac{6}{7} (2x^4 + x^3) dx \\
&= \frac{12}{35} + \frac{3}{14} = \frac{39}{70}
\end{aligned}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \frac{39}{70} - \frac{25}{49} = \frac{23}{490}$$

$$\begin{aligned}
E(Y^2) &= \int_{-\infty}^{\infty} y^2 \int_{-\infty}^{\infty} p(x, y) dx dy \\
&= \int_0^1 \int_0^2 \frac{6}{7} \left(x^2 y^2 + \frac{1}{2} x y^3 \right) dy dx \\
&= \int_0^1 \frac{6}{7} \left(\frac{8}{3} x^2 + 2x \right) dx \\
&= \frac{34}{21}
\end{aligned}$$

故

$$\text{Var}(Y) = E(Y^2) - (E(Y))^2 = \frac{34}{21} - \frac{64}{49} = \frac{46}{147}$$

因此

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}} = -\frac{\sqrt{5}}{23\sqrt{3}} = -\frac{\sqrt{15}}{69}$$

◇

Exercise 12.11

设二维随机变量 (X, Y) 在矩形

$$G = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 1\}$$

上服从均匀分布, 记

$$U = \begin{cases} 1, & X > Y, \\ 0, & X \leq Y, \end{cases} \quad V = \begin{cases} 1, & X > 2Y, \\ 0, & X \leq 2Y. \end{cases}$$

求 U 和 V 的相关系数。

Solution. 注意到 $UV = V$, 我们有

$$\text{Cov}(U, V) = E(UV) - E(U)E(V) = E(V)(1 - E(U))$$

其中

$$E(U) = \int \chi_{\{X > Y\}} dP = \int_0^1 \int_y^2 \frac{1}{2} dx dy = \int_0^1 \frac{1}{2} (2 - y) dy = \frac{3}{4}$$

$$E(V) = \int \chi_{\{X > 2Y\}} dP = \int_0^1 \int_{2y}^2 \frac{1}{2} dx dy = \int_0^1 \frac{1}{2} (2 - 2y) dy = \frac{1}{2}$$

因此

$$\text{Cov}(U, V) = \frac{1}{2} \left(1 - \frac{3}{4} \right) = \frac{1}{8}$$

$$E(U^2) = \int \chi_{\{X > Y\}} dP = \frac{3}{4}$$

$$E(V^2) = \int \chi_{\{X > 2Y\}} dP = \frac{1}{2}$$

$$\text{Var}(U) = E(U^2) - (E(U))^2 = \frac{3}{16}$$

$$\text{Var}(V) = E(V^2) - (E(V))^2 = \frac{1}{4}$$

于是

$$\text{Corr}(U, V) = \frac{\text{Cov}(U, V)}{\sqrt{\text{Var}(U)\text{Var}(V)}} = \frac{1}{8} \sqrt{\frac{16}{3}} = \frac{1}{\sqrt{3}}$$

◇

Exercise 12.12

设 a 为区间 $(0,1)$ 上的一个定点, 随机变量 X 服从区间 $(0,1)$ 上的均匀分布, 以 Y 表示点 X 到 a 的距离, 问 a 为何值时 X 与 Y 不相关.

Solution. $Y = |X - a| = \chi_{\{X > a\}}(X - a) + \chi_{\{X \leq a\}}(a - X).$

$$\begin{aligned} E(XY) &= \int XY \, dP \\ &= \int_a^1 x(x-a) \, dx + \int_0^a x(a-x) \, dx \\ &= \frac{1}{3}(1-a^3) - \frac{1}{2}a(1-a^2) + \frac{1}{2}a^3 - \frac{1}{3}a^3 \\ &= \frac{1}{3}a^3 - \frac{1}{2}a + \frac{1}{3} \end{aligned}$$

$$E(X) = \int_0^1 x \, dx = \frac{1}{2}$$

$$\begin{aligned} E(Y) &= \int_a^1 (x-a) \, dx + \int_0^a (a-x) \, dx \\ &= \frac{1}{2}(1-a^2) - a(1-a) + a^2 - \frac{1}{2}a^2 \\ &= a^2 - a + \frac{1}{2} \end{aligned}$$

因此

$$\begin{aligned} \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) = \frac{1}{3}a^3 - \frac{1}{2}a + \frac{1}{3} - \frac{1}{2}\left(a^2 - a + \frac{1}{2}\right) \\ &= \frac{1}{3}a^3 - \frac{1}{2}a^2 + \frac{1}{12} \end{aligned}$$

X, Y 不相关, 当且仅当 $\text{Cov}(X, Y) = 0$, 当且仅当

$$\frac{1}{3}a^3 - \frac{1}{2}a^2 + \frac{1}{12} = 0, \quad a \in (0, 1)$$

解方程得 $4a^3 - 6a^2 + 1 = (2a - 1)(2a^2 - 2a - 1)$

$$2a^2 - 2a - 1 = 0 \iff a = \frac{1 \pm \sqrt{3}}{2} \notin (0, 1)$$

因此只有 $a = \frac{1}{2}$.

◇

Exercise 12.13

设随机向量 (X_1, X_2, X_3) 满足条件

$$aX_1 + bX_2 + cX_3 = 0,$$

$$E(X_1) = E(X_2) = E(X_3) = d,$$

$$\text{Var}(X_1) = \text{Var}(X_2) = \text{Var}(X_3) = \sigma^2.$$

其中 a, b, c, d, σ^2 均为常数, 求相关系数 $\rho_{12}, \rho_{23}, \rho_{13}$.

Proof. 由

$$-cX_3 = aX_1 + bX_2$$

得到

$$\begin{aligned} c^2\sigma^2 &= \text{Var}(cX_3) = \text{Var}(aX_1 + bX_2) \\ &= \text{Cov}(aX_1 + bX_2, aX_1 + bX_2) \\ &= a^2\text{Var}(X_1) + b^2\text{Var}(X_2) + 2ab\sqrt{\text{Var}(X_1)\text{Var}(X_2)}\rho_{12} \\ &= (a^2 + b^2)\sigma^2 + 2ab\sigma^2\rho_{12} \end{aligned}$$

从而

$$\rho_{12} = \frac{c^2 - a^2 - b^2}{2ab}$$

由对称性,

$$\rho_{23} = \frac{a^2 - b^2 - c^2}{2bc}$$

$$\rho_{13} = \frac{b^2 - c^2 - a^2}{2ac}$$

□

Exercise 12.14

设随机变量 X 服从区间 $(-0.5, 0.5)$ 上的均匀分布, $Y = \cos X$, 则 X 与 Y 有函数关系. 试求 $\text{Cov}(X, Y)$.

Solution.

$$\begin{aligned} E(XY) &= \int X \cos X \, dP \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} x \cos x \, dx \\ &= 0 \end{aligned}$$

$$E(X) = \int X \, dP = \int_{-\frac{1}{2}}^{\frac{1}{2}} x \, dx = 0$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0$$

◇

Exercise 12.15

设二维随机变量 (X, Y) 服从单位圆内的均匀分布, 其联合密度函数为

$$\rho(x, y) = \begin{cases} \frac{1}{\pi}, & x^2 + y^2 < 1, \\ 0, & x^2 + y^2 \geq 1. \end{cases}$$

试证 X 与 Y 不独立且 X 与 Y 不相关.

Proof.

$$\rho_X(x) = \int_{-\infty}^{\infty} \frac{1}{\pi} \chi_{\{x^2+y^2<1\}} \, dy = \chi_{\{-1<x<1\}} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} \, dy = \chi_{\{-1<x<1\}} \frac{2\sqrt{1-x^2}}{\pi}$$

由对称性

$$\rho_Y(y) = \chi_{\{-1<y<1\}} \frac{2\sqrt{1-y^2}}{\pi}$$

得到

$$\rho_X(x)\rho_Y(y) = \chi_{\{-1 < x, y < 1\}} \frac{4\sqrt{(1-x^2)(1-y^2)}}{\pi^2}$$

由于函数有不同的支撑集, $\rho_X(x)\rho_Y(y) \neq \rho_{X,Y}(x,y)$, X, Y 不独立.

$$\begin{aligned} E(XY) &= \iint \chi_{\{x^2+y^2 < 1\}} \frac{1}{\pi} xy \, dx \, dy \\ &= \iint \chi_{\{r < 1\}} \frac{1}{\pi} r^2 \sin \theta \cos \theta r \, dr \, d\theta \\ &= \int_0^1 \int_0^{2\pi} \frac{1}{2\pi} r^3 \sin 2\theta \, dr \, d\theta \\ &= 0 \end{aligned}$$

$$E(X) = \int_{-1}^1 \frac{2\sqrt{1-x^2}}{\pi} x \, dx = 0$$

其中等号成立是因为被积函数是奇函数, 且积分区域关于 0 对称. 因此

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0 - 0 = 0$$

因此 X, Y 不相关. □

Exercise 12.16

设随机向量 (X_1, X_2, X_3) 的相关系数分别为 $\rho_{12}, \rho_{23}, \rho_{31}$, 且

$$E(X_1) = E(X_2) = E(X_3) = 0, \quad \text{Var}(X_1) = \text{Var}(X_2) = \text{Var}(X_3) = \sigma^2,$$

令

$$Y_1 = X_1 + X_2, \quad Y_2 = X_2 + X_3, \quad Y_3 = X_3 + X_1.$$

证明: Y_1, Y_2, Y_3 两两不相关的充要条件为 $\rho_{12} + \rho_{23} + \rho_{31} = -1$.

Proof. 只考虑 $\sigma > 0$, 若不然 $X_1 = X_2 = X_3 = 0$, P -a.e.

若 $\rho_{12} + \rho_{23} + \rho_{31} = -1$, 则

$$\text{Cov}(Y_1, Y_2) = \text{Cov}(X_1 + X_2, X_2 + X_3) = \sigma^2 + \sigma^2 \rho_{12} + \sigma^2 \rho_{23} + \sigma^2 \rho_{13} = \sigma^2 - \sigma^2 = 0$$

由对称性

$$\text{Cov}(Y_2, Y_3) = \text{Cov}(Y_1, Y_3) = 0$$

反之, 若 Y_1, Y_2, Y_3 两两不相关, 则特别地

$$0 = \text{Cov}(Y_1, Y_2) = \sigma^2(\rho_{12} + \rho_{23} + \rho_{13} + 1)$$

因此

$$\rho_{12} + \rho_{23} + \rho_{13} = -1$$

□

Exercise 12.17

设二维随机变量 (X, Y) 的联合密度函数如下, 试求 (X, Y) 的协方差矩阵:

1.

$$p_1(x, y) = \begin{cases} 6xy^2, & 0 < x < 1, \quad 0 < y < 1, \\ 0, & \text{其他.} \end{cases}$$

2.

$$p_2(x, y) = \begin{cases} \frac{x+y}{8}, & 0 < x < 2, \quad 0 < y < 2, \\ 0, & \text{其他.} \end{cases}$$

Solution. 1.

$$f_X(x) = \int_0^1 6xy^2 dy = 6x \int_0^1 y^2 dy = 6x \cdot \frac{1}{3} = 2x, \quad 0 < x < 1,$$

$$f_Y(y) = \int_0^1 6xy^2 dx = 6y^2 \int_0^1 x dx = 6y^2 \cdot \frac{1}{2} = 3y^2, \quad 0 < y < 1.$$

注意到

$$f_X(x)f_Y(y) = 2x \cdot 3y^2 = 6xy^2 = p_1(x, y),$$

所以 X, Y 相互独立, 因此 $\text{Cov}(X, Y) = 0$.

$$E[X] = \int_0^1 xf_X(x) dx = \int_0^1 x \cdot 2x dx = 2 \int_0^1 x^2 dx = 2 \cdot \frac{1}{3} = \frac{2}{3},$$

$$E[X^2] = \int_0^1 x^2 f_X(x) dx = \int_0^1 x^2 \cdot 2x dx = 2 \int_0^1 x^3 dx = 2 \cdot \frac{1}{4} = \frac{1}{2},$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{18}.$$

$$E[Y] = \int_0^1 y f_Y(y) dy = \int_0^1 y \cdot 3y^2 dy = 3 \int_0^1 y^3 dy = 3 \cdot \frac{1}{4} = \frac{3}{4},$$

$$E[Y^2] = \int_0^1 y^2 f_Y(y) dy = \int_0^1 y^2 \cdot 3y^2 dy = 3 \int_0^1 y^4 dy = 3 \cdot \frac{1}{5} = \frac{3}{5},$$

$$\text{Var}(Y) = E[Y^2] - E[Y]^2 = \frac{3}{5} - \left(\frac{3}{4}\right)^2 = \frac{3}{5} - \frac{9}{16} = \frac{3}{80}.$$

协方差矩阵为

$$\Sigma_1 = \begin{pmatrix} \frac{1}{18} & 0 \\ 0 & \frac{3}{80} \end{pmatrix}.$$

2.

$$\begin{aligned} f_X(x) &= \int_0^2 \frac{x+y}{8} dy = \frac{1}{8} \left(\int_0^2 x dy + \int_0^2 y dy \right) \\ &= \frac{1}{8} \left(2x + \frac{2^2}{2} \right) = \frac{1}{8} (2x + 2) = \frac{x+1}{4}, \quad 0 < x < 2, \end{aligned}$$

$$f_Y(y) = \frac{y+1}{4}, \quad 0 < y < 2.$$

由对称性可知 $E[X] = E[Y]$, $\text{Var}(X) = \text{Var}(Y)$.

$$\begin{aligned} E[X] &= \int_0^2 x f_X(x) dx = \int_0^2 x \cdot \frac{x+1}{4} dx = \frac{1}{4} \int_0^2 (x^2 + x) dx \\ &= \frac{1}{4} \left(\frac{2^3}{3} + \frac{2^2}{2} \right) = \frac{1}{4} \left(\frac{8}{3} + 2 \right) = \frac{1}{4} \cdot \frac{14}{3} = \frac{7}{6}, \end{aligned}$$

$$\begin{aligned} E[X^2] &= \int_0^2 x^2 f_X(x) dx = \int_0^2 x^2 \cdot \frac{x+1}{4} dx = \frac{1}{4} \int_0^2 (x^3 + x^2) dx \\ &= \frac{1}{4} \left(\frac{2^4}{4} + \frac{2^3}{3} \right) = \frac{1}{4} \left(4 + \frac{8}{3} \right) = \frac{1}{4} \cdot \frac{20}{3} = \frac{5}{3}, \end{aligned}$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = \frac{5}{3} - \left(\frac{7}{6}\right)^2 = \frac{5}{3} - \frac{49}{36} = \frac{60 - 49}{36} = \frac{11}{36}.$$

因此 $\text{Var}(Y) = \frac{11}{36}$.

$$\begin{aligned} E[XY] &= \int_0^2 \int_0^2 xy \frac{x+y}{8} dx dy \\ &= \frac{1}{8} \int_0^2 \int_0^2 (x^2y + xy^2) dx dy. \end{aligned}$$

两项对称, 先算 $\int_0^2 \int_0^2 x^2y dx dy$.

$$\begin{aligned} \int_0^2 \int_0^2 x^2y dx dy &= \int_0^2 \left(y \int_0^2 x^2 dx \right) dy = \int_0^2 y \cdot \frac{2^3}{3} dy = \frac{8}{3} \int_0^2 y dy \\ &= \frac{8}{3} \cdot \frac{2^2}{2} = \frac{8}{3} \cdot 2 = \frac{16}{3}. \end{aligned}$$

同理 $\int_0^2 \int_0^2 xy^2 dx dy = \frac{16}{3}$.

所以

$$E[XY] = \frac{1}{8} \left(\frac{16}{3} + \frac{16}{3} \right) = \frac{1}{8} \cdot \frac{32}{3} = \frac{4}{3}.$$

于是

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = \frac{4}{3} - \left(\frac{7}{6}\right)^2 = \frac{4}{3} - \frac{49}{36} = \frac{48 - 49}{36} = -\frac{1}{36}.$$

协方差矩阵为

$$\Sigma_2 = \begin{pmatrix} \frac{11}{36} & -\frac{1}{36} \\ -\frac{1}{36} & \frac{11}{36} \end{pmatrix}.$$

◇

Exercise 12.18

设随机变量 X 与 Y 都只能取两个值, 试证: X 与 Y 独立与不相关是等价的.

Solution. 设 X 的取值为 x_1, x_2 , 设 $A = \{X = x_1\}, B = \{Y = y_1\}$, 则

$$X = x_2 + (x_1 - x_2)\chi_A, \quad Y = y_2 + (y_1 - y_2)\chi_B$$

则

$$E(X) = x_2 + (x_1 - x_2)P(A)$$

$$E(Y) = y_2 + (y_1 - y_2)P(B)$$

$$XY = x_2y_2 + x_2(y_1 - y_2)\chi_B + y_2(x_1 - x_2)\chi_A + (x_1 - x_2)(y_1 - y_2)\chi_A\chi_B$$

$$E(XY) = x_2y_2 + x_2(y_1 - y_2)P(B) + y_2(x_1 - x_2)P(A) + (x_1 - x_2)(y_1 - y_2)P(A \cap B)$$

因此

$$E(XY) - E(X)E(Y) = (x_1 - x_2)(y_1 - y_2)(P(A \cap B) - P(A)P(B))$$

若 $x_1 = x_2$, 则 X 是常值的, 此时总有 X 与 Y 独立且不相关. 若 $y_1 = y_2$ 也有类似的结论成立.

若 $x_1 \neq x_2$ 且 $y_1 \neq y_2$, 则

$$X, Y \text{ 不相关} \iff \text{Cov}(X, Y) = 0 \iff P(A)P(B) = P(A \cap B) \iff X, Y \text{ 独立}$$

◇

Exercise 12.19

设随机向量 (X_1, X_2, X_3) 的相关系数分别为 $\rho_{12}, \rho_{23}, \rho_{31}$, 证明

$$\rho_{12}^2 + \rho_{23}^2 + \rho_{31}^2 \leq 1 + 2\rho_{12}\rho_{23}\rho_{31}.$$

Proof. 先考虑 $E(X_1) = E(X_2) = E(X_3) = 0, \text{Var}(X_1) = \text{Var}(X_2) = \text{Var}(X_3) = 1$ 的情况. 则此时

$$\text{Cov}(aX_1 + bX_2 + cX_3) = (a, b, c) \begin{pmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

由于对于任意的 a, b, c , $\text{Cov}(aX_1 + bX_2 + cX_3) \geq 0$, 我们有右侧的矩阵是半正定的, 从而

$$1 - \rho_{12}^2 + \rho_{23}^2 + \rho_{31}^2 - 2\rho_{12}\rho_{23}\rho_{31} = \det \begin{pmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{pmatrix} \geq 0$$

对于一般的情况, 令

$$Y_i = \frac{X_i - E(X_i)}{\sqrt{\text{Var}(X_i)}}$$

则 $E(Y_i) = 0, \text{Var}(Y_i) = 1, i = 1, 2, 3$. 记 $\rho'_{ij} = \text{Cov}(Y_i, Y_j)$, 则

$$\rho'_{ij} = \frac{1}{\sqrt{\text{Var}(X_i) \text{Var}(X_j)}} \text{Cov}(X_i, X_j) = \rho_{ij}$$

之前的讨论说明了所需不等式对 ρ'_{ij} 成立, 从而 ρ_{ij} 满足所需不等式. □

Exercise 12.20

设随机变量 $X \sim N(0, 1)$, Y 各以 0.5 的概率取值 ± 1 , 且假定 X 与 Y 相互独立. 令 $Z = X \cdot Y$, 证明:

1. $Z \sim N(0, 1)$;
2. X 与 Z 不相关, 但不独立.

Proof. 1.

$$\begin{aligned} P(Z \leq k) &= P(X \cdot Y \leq k) \\ &= P(X \leq k, Y = 1) + P(X \geq -k, Y = -1) \\ &= \frac{1}{2}P(X \leq k) + \frac{1}{2}P(X \geq -k) \\ &= \frac{1}{2}P(X \leq k) + \frac{1}{2}P(X \leq k) \\ &= P(X \leq k) \end{aligned}$$

因此 Z 与 X 同分布, $Z \sim N(0, 1)$.

2.

$$\begin{aligned} P(X^2 Y \leq k) &= P(X^2 \leq k, Y = 1) + P(X^2 \geq -k, Y = -1) \\ &= \frac{1}{2}P(X^2 \leq k) + \frac{1}{2}P(X^2 \geq -k) \end{aligned}$$

此外

$$P(X^2 Y \geq k) = \frac{1}{2}P(X^2 \geq k) + \frac{1}{2}P(X^2 \leq -k)$$

$$\begin{aligned}
 E(XZ) &= E(X^2Y) = \int_0^{\infty} P(X^2Y \geq x) - P(X^2Y \leq -x) dx \\
 &= \frac{1}{2} \int_0^{\infty} P(X^2 \geq x) + P(X^2 \leq -x) - P(X^2 \leq -x) - P(X^2 \geq x) dx \\
 &= 0
 \end{aligned}$$

又

$$E(X) = E(Z) = 0$$

因此

$$\text{Cov}(X, Z) = E(XZ) - E(X)E(Z) = 0$$

故 X, Z 不相关.

对于 $k > 0$, 我们有

$$\begin{aligned}
 P(X \leq -k, Z \leq -k) &= P(X \leq -k, XY \leq -k) \\
 &= P(X \leq -k, XY \leq -k, Y = 1) + P(X \leq -k, XY \leq -k, Y = -1) \\
 &= \frac{1}{2}P(X \leq -k) + \frac{1}{2}P(k \leq X \leq -k) \\
 &= \frac{1}{2}P(X \leq -k)
 \end{aligned}$$

而 $P(X \leq -k) \neq 0, P(Z \leq -k) < \frac{1}{2}$, 因此

$$\frac{1}{2}P(X \leq -k) \neq P(X \leq -k)P(Z \leq -k)$$

从而

$$P(X \leq -k, Z \leq -k) \neq P(X \leq -k)P(Z \leq -k)$$

因此 X, Z 不独立.

□

Exercise 12.21

设随机向量 (X, Y) 满足

$$E(X) = E(Y) = 0, \quad \text{Var}(X) = \text{Var}(Y) = 1, \quad \text{Cov}(X, Y) = \rho.$$

证明:

$$E(\max\{X^2, Y^2\}) \leq 1 + \sqrt{1 - \rho^2}.$$

Proof.

$$\max\{X^2, Y^2\} = \frac{X^2 + Y^2 + |X^2 - Y^2|}{2}$$

$$E(X^2) = \text{Var}(X) - (E(X))^2 = 1$$

因此

$$E(\max\{X^2, Y^2\}) = E\left(\frac{X^2 + Y^2 + |X^2 - Y^2|}{2}\right) = 1 + \frac{1}{2}E(|X^2 - Y^2|)$$

$$E(|X^2 - Y^2|) = E(|X - Y| |X + Y|) \leq \sqrt{E(|X - Y|^2) E(|X + Y|^2)}$$

由于 $E(X - Y) = E(X) - E(Y) = 0$, $E(X + Y) = E(X) + E(Y) = 0$.

$$\begin{aligned} E((X - Y)^2) &= \text{Var}((X - Y)) \\ &= \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y) \\ &= 2 - 2\rho \end{aligned}$$

$$\begin{aligned} E((X + Y)^2) &= \text{Var}((X + Y)) \\ &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \\ &= 2 + 2\rho \end{aligned}$$

因此

$$E(|X^2 - Y^2|) \leq \sqrt{(2 - 2\rho)(2 + \rho)} = 2\sqrt{1 - \rho^2}$$

$$E(\max\{X^2, Y^2\}) \leq 1 + \sqrt{1 - \rho^2}$$

□

Exercise 12.22

设随机变量 $X_1, X_2, \dots, X_n$ 中任意两个的相关系数都是 ρ , 试证:

$$\rho \geq -\frac{1}{n-1}.$$

Proof. 令

$$Y_i = \frac{X_i - E(X_i)}{\sqrt{\text{Var}(X_i)}}$$

则对于 $i \neq j$,

$$\begin{aligned} \text{Cov}(Y_i, Y_j) &= \text{Cov}\left(\frac{X_i - E(X_i)}{\sqrt{\text{Var}(X_i)}}, \frac{X_j - E(X_j)}{\sqrt{\text{Var}(X_j)}}\right) \\ &= \frac{1}{\sqrt{\text{Var}(X_i)}\sqrt{\text{Var}(X_j)}} \text{Cov}(X_i, X_j) \\ &= \text{Corr}(X_i, X_j) = \rho \end{aligned}$$

于是

$$\begin{aligned} 0 \leq \text{Cov}(Y_1 + \cdots + Y_n, Y_1 + \cdots + Y_n) &= \sum_{1 \leq i, j \leq n} \text{Cov}(Y_i, Y_j) \\ &= \sum_{i=j} \text{Cov}(Y_i, Y_i) + \sum_{i \neq j} \text{Cov}(Y_i, Y_j) \\ &= n + (n^2 - n)\rho \end{aligned}$$

因此

$$\rho \geq \frac{-n}{n^2 - n} = -\frac{1}{n-1}$$

□

Exercise 12.23

已知 (X, Y) 的联合分布列如下

$$P(X = 1, Y = 1) = P(X = 2, Y = 1) = \frac{1}{8},$$

$$P(X = 1, Y = 2) = \frac{1}{4}, \quad P(X = 2, Y = 2) = \frac{1}{2}.$$

试求

1. 已知 $Y = i$ 的条件下, X 的条件分布列, $i = 1, 2$;
2. X 与 Y 是否独立?

Proof. 1.

$$P(Y = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$P(Y = 2) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$$P(X = 1|Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)} = \frac{1}{2}$$

$$P(X = 2|Y = 1) = \frac{P(X = 2, Y = 1)}{P(Y = 1)} = \frac{1}{2}$$

$$P(X = 1|Y = 2) = \frac{P(X = 1, Y = 2)}{P(Y = 2)} = \frac{1}{3}$$

$$P(X = 2|Y = 2) = \frac{P(X = 2, Y = 2)}{P(Y = 2)} = \frac{2}{3}$$

2.

$$P(X = 1) = \frac{3}{8}, \quad P(X = 2) = \frac{5}{8}$$

因此

$$P(X = 1)P(Y = 1) = \frac{3}{32} \neq \frac{1}{8} = P(X = 1, Y = 1)$$

因此 X, Y 不独立.

□

Exercise 12.24

设随机变量 X 与 Y 独立同分布, 试在以下情况下求 $P(X = k | X + Y = m)$:

1. X 与 Y 都服从参数为 p 的几何分布;
2. X 与 Y 都服从参数为 (n, p) 的二项分布.

Proof. 1.

$$P(X = k|X + Y = m) = \frac{P(X = k, X + Y = m)}{P(X + Y = m)} = \frac{P(X = k, Y = m - k)}{P(X + Y = m)}$$

其中

$$\begin{aligned}
 P(X + Y = m) &= \sum_{k=1}^{m-1} P(X = k) P(Y = m - k) \\
 &= \sum_{k=1}^{m-1} (1 - p)^{k-1} p (1 - p)^{m-k-1} p \\
 &= \sum_{k=1}^{m-1} (1 - p)^{m-2} p^2 \\
 &= (m - 1) (1 - p)^{m-2} p^2
 \end{aligned}$$

$$\begin{aligned}
 P(X = k, Y = m - k) &= P(X = k) P(Y = m - k) \\
 &= (1 - p)^{k-1} p (1 - p)^{m-k-1} p \\
 &= (1 - p)^{m-2} p^2
 \end{aligned}$$

故

$$P(X = k | X + Y = m) = \frac{(1 - p)^{m-2} p^2}{(m - 1) (1 - p)^{m-2} p^2} = \frac{1}{m - 1}$$

2.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\begin{aligned}
 P(X + Y = m) &= \sum_{k=0}^m P(X = k) P(Y = m - k) \\
 &= \sum_{k=0}^m \binom{n}{k} p^k (1 - p)^{n-k} \binom{n}{m-k} p^{m-k} (1 - p)^{n-m+k} \\
 &= \sum_{k=0}^m \binom{n}{k} \binom{n}{m-k} p^m (1 - p)^{2n-m} \\
 &= \binom{2n}{m} p^m (1 - p)^{2n-m}
 \end{aligned}$$

因此

$$\begin{aligned}
 P(X = k | X + Y = m) &= \frac{P(X = k, Y = m - k)}{P(X + Y = m)} \\
 &= \frac{\binom{n}{k} p^k (1-p)^{n-k} \binom{n}{m-k} p^{m-k} (1-p)^{n-m+k}}{\binom{2n}{m} p^m (1-p)^{2n-m}} \\
 &= \frac{\binom{n}{k} \binom{n}{m-k} p^m (1-p)^{2n-m}}{\binom{2n}{m} p^m (1-p)^{2n-m}} \\
 &= \frac{\binom{n}{k} \binom{n}{m-k}}{\binom{2n}{m}}
 \end{aligned}$$

□

Exercise 12.25

设二维连续随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} 3x, & 0 < x < 1, 0 < y < x, \\ 0, & \text{其他.} \end{cases}$$

试求条件密度函数 $p(y | x)$.

Proof.

$$p(y|x) = \frac{p(x, y)}{p_X(x)}$$

其中

$$p_X(x) = \chi_{\{0 < x < 1\}} \int_0^x 3x \, dy = \chi_{\{0 < x < 1\}} 3x^2$$

因此

$$p(y|x) = \chi_{\{0 < y < x < 1\}} \frac{1}{x}$$

□

Exercise 12.26

设二维连续随机变量 (X, Y) 的联合密度函数为

$$p(x, y) = \begin{cases} \frac{21}{4}x^2y, & x^2 \leq y \leq 1, \\ 0, & \text{其他.} \end{cases}$$

求条件概率 $P(Y \geq 0.75 \mid X = 0.5)$.

Proof.

$$p_{Y|X}(y|0.5) = \frac{p(x, y)}{p_X(0.5)}$$

其中

$$p_X(0.5) = \int_{\frac{1}{4}}^1 \frac{21}{4} \frac{1}{4} y \, dy = \frac{21}{16} \frac{1}{2} \left(1 - \frac{1}{16}\right) = \frac{21 \times 15}{32 \times 16}$$

于是

$$p_{Y|X}(y|0.5) = \chi_{\{\frac{1}{4} \leq y \leq 1\}} \frac{32}{15} y$$

$$P(Y \geq 0.75 \mid X = 0.5) = \int_{\frac{3}{4}}^1 \frac{32}{15} y \, dy = \frac{16}{15} \left(1 - \frac{9}{16}\right) = \frac{7}{15}$$

□